

Law and Economics in the Age of Computers

What are the implications of computational complexity for law and economics?

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2026-09-18

Yale, Columbia and NBER

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Introduction

Helped

- The law and economics movement has help to organize and improve the law in a wide variety of areas.
- For the most part this has relied on price theory, but there are a number of questions that price theory cannot fully address.
- This lecture breaks down the analytical tools from economics into three distinct class of models:
 - Price theory
 - Information economics
 - Bounded Rationality

Goals of this Lecture I

- Each class of models provides a different lens through which to analyze legal and economic issues.
- The choice of model depends on the specific legal and economic issue being analyzed, and whether the framework can lead to better decisions.
- Discuss the implications of bounded rationality models, in particular they rely on the existence of explicit values.
- Relate this result on HLA Hart's work on legal realism and the "penumbra of doubt".
- Final point is that this leads to an agenda for law and economics scholars to contribute to AI regulation using the insights from well functioning legal system. ∴ {notes} ∴

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- All complex human societies have developed legal systems to regulate behavior, resolve disputes, and maintain social order.
- These systems have evolved over time to address the changing needs and complexities of human societies, and hence be can viewed, in the sense of Posner (1973) as improving social outcomes over time. **can be**

Hammurabi's Code



- Hammurabi's Code is one of the earliest legal codes, dating back to around 1750 BC in ancient Mesopotamia. It is a surprisingly comprehensive set of laws that covers various aspects of society, including family, property, immorality, theft, burglary, wage and price regulation, and even professional conduct.
- See Vincent (1904) for comprehensive discussion of the code.

Implications of Hammurabi's Code for Today

- The code is famously known for its principle of “an eye for an eye,” which emphasizes the idea of retributive justice.
- It also includes provisions for social hierarchy, with different punishments based on the social status of the offender and the victim.
- What is interesting is that many of the main issues have not changed in 3,778 years!
- What has changed are:
 - New technologies that requires new notions of property rights.
 - Better economic theory and evidence to guide the design of better rules (the goal of this course).

Law and Economics: Old School and New School Economics

- Law and economics movement started with the founding of the *Journal of Law and Economics* by Aaron Director and Ronald Coase at the University of Chicago.
- The textbook Posner (2014) was very influential in bringing price theory to the law school.

Note use of the citation key results in a citation rather than the full name of the textbook

The original work in law and economics was very much *old school* economics (aka “Chicago School of Law and Economics”)

- The economy is viewed from the perspective of *price theory*: free markets are the most efficient way to allocate resources.
- Posner argued that legal rules can be explained as (imperfect) wealth maximizing solutions, that over time lead to more efficient markets.
- The main barriers to the efficient operation of the economy are government overreach and monopolies, such as unions and monopolized industries. **(according to Posner)**

Old school is necessary (very useful) but not sufficient to address many questions in law and economics.

- Price theory is a good first order approach to understanding both the economy and the law.
- In addition to Posner's book, both Cooter and Ulen (2016) and Shavell (2004) provide excellent treatments of law and economics.
- These books use the intuition of price theory, combined with game theory (see **Broken formatting on BairdGertner, probably due to the line break in the code here** 1) and transactions cost economics (Williamson (1985)) to provide a framework for the study of legal rules.

New school economics builds on contributions of a number of
Nobel prize winners

New School Economics I: Failures of Price Theory due to Imperfect Information i

Should 2007 be moved to the front? Division may be confusing before outlining the distinction

Year	Laureates	Contribution Summary
2001	George A. Akerlof, A. Michael Spence, Joseph E. Stiglitz	For fundamental analyses of markets with asymmetric information (when buyers or sellers have unequal information).
2010	Peter A. Diamond, Dale T. Mortensen, Christopher A. Pissarides	For analyzing markets with search frictions (helping explain why high unemployment can coexist with job vacancies).
2016	Oliver Hart, Bengt Holmström	For foundational contributions to Contract Theory (analyzing executive pay, performance contracts, and privatization).
2007	Leonid Hurwicz, Eric S. Maskin, Roger B. Myerson	For establishing Mechanism Design Theory (how institutions achieve desirable economic outcomes when information is private).

Year	Laureates	Contribution Summary
2002	Daniel Kahneman, Vernon L. Smith	For integrating psychological research into economic decision-making under uncertainty (Kahneman) and establishing empirical laboratory experiments (Smith).
2017	Richard Thaler	For incorporating psychologically realistic assumptions into economic decision-making (Behavioral Economics).

New school Economics III: Analysis of Big Data (Causal and Inductive) i

Year	Laureates	Contribution Summary
2021	David Card, Joshua Angrist, Guido Imbens	For empirical contributions to labor economics (Card) and methodological breakthroughs in analyzing causal relationships using natural experiments (Angrist/Imbens).
2024	John J. Hopfield & Geoffrey E. Hinton	Foundational Neural Network Architectures: Awarded for foundational discoveries that enable machine learning with artificial neural networks. Hopfield created an associative memory (Hopfield Network) that can store and reconstruct patterns. Hinton built on this to invent the Boltzmann machine, paving the way for deep learning architectures that underpin modern generative AI models.

2024	Demis Hassabis & John M. Jumper (shared with David Baker)	AI for Protein Structure Prediction: Awarded for developing AlphaFold2 , an AI system that solved a 50-year-old grand challenge in biology by predicting 3D structures for virtually all known proteins from their amino acid sequences. This revolutionized molecular biology, drug discovery, and biochemical research.
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For the purposes of this lecture, New School Economics is divided into three groups:

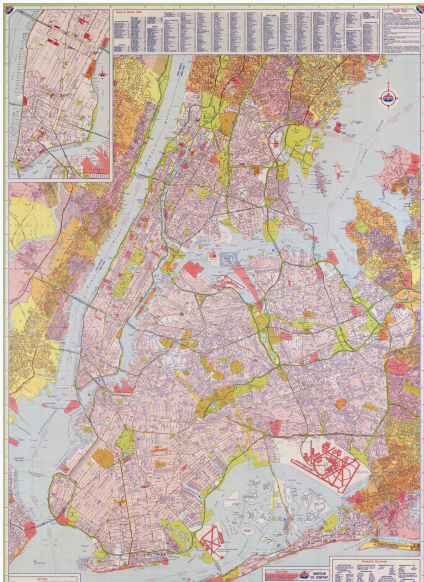
- Price Theory and Game Theory (1940-1980) - the foundations of modern economics (appendix MacLeod (2022)). **This could probably go before the timeline**
- Information Economics, Contract Theory and Causal Analysis (1980-2010)
- Bounded Rationality, Behavioral Economics and Big Data (2010-now)

“Not only is it not right, it is not even wrong.” — Wolfgang Pauli
(Peierls (1960))

Why do we build models that clearly wrong?

- One of the most common **misunderstandings** regarding science is that the goal is to find the “true” model of the world. **This should probably be explained, i.e., no model is correct**
- Models are simplified representations of reality that are *decision-making aids or tools*.
- This is certainly true for physics, where the choice of model depends upon the problem, time and distance scales — from the Planck scales of 10^{-43} seconds (the Planck time) and lengths around 10^{-35} meters (the Planck length) to the cosmological scales where the age of the universe is 10^{10} years and has a diameter of around 10^{10} light years.

Example — 1963 Map of New York



- This map provides a lot of information about New York City, including the location of streets, parks, and buildings.
- While you can use it to navigate the subway system, it is “difficult” to use since the print is small, making it hard to find the subway stations.

Not only because the print is small but because subway stations themselves are not laid out in an easily comprehensible manner

Example — Subway Map of New York

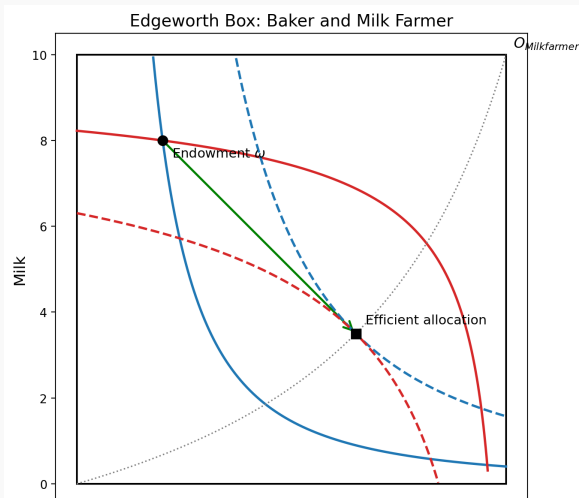


- This is as “wrong” map, but provides a much more useful guide to choosing a subway route. **map but**
- This is a feature of *all* models - they are simplifications of the real world that allow humans to make better decisions.

Standard Economic Models

Price Theory (see appendix of MacLeod (2022))

- When we add prices and require supply=demand, then for any endowment there is a *unique* market allocation.



Price theory is useful, even though it is “wrong”!

- Price theory provides a framework for understanding the allocation of commodities when they are well defined.
- It provides solutions for a number of important policy issues, including pollution (via taxes or tradable permits), congestion pricing (as in London and NYC), and the allocation of property rights via auctions (as in the FCC spectrum auctions, or timber auctions).
- It is the basis for antitrust policy and the control of monopoly power. The extent to which monopoly power is an efficiency or equity issue depends on the nature of information—the topic of today’s lecture.

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- Model building algorithm:
 - What are the relevant states of Nature (Savage (1972 (first published 1954)))
 - What are the actions and payoffs available to the decision makers.
 - What are the beliefs of the decision makers over states and action.
 - Individuals choose Markov perfect strategies.

This theory has been very successful in extending the range of economic analysis.

- Milgrom and Roberts (1992) - Amazing review of how information economics explains organizations.
Again, strange formatting with BairdGertner, also note Harvard is misspelled in Zotero entry for BairdGertner
- 1 - Use information economics to explain many legal rules.
- Gennaioli and Shleifer (2007) - models the evolution of law.
- Listokin (2010) - uses Bayesian methods to set contract damages.
- Chakravarty and MacLeod (2009) - uses information economics to explain the structure of construction contracts.
- Many papers at this conference!

Economics for the Age of Computers - Bounded Rationality and Computational Complexity

Bounded Rationality and Computational Complexity

- Bounded rationality recognizes that individuals have limited cognitive resources and cannot always make fully rational decisions.
- Computational complexity considers the difficulty of solving problems and making decisions given these cognitive limitations. **I start thinking of heuristics and priors: Limited time to compute**
So maybe the “Bounded rationality” and “Computational complexity” points should be flipped
- Together, these concepts help explain why real-world decision-making often deviates from the predictions of standard economic models.

A Cognitive Model of Bounded Rationality (Simon (1955), MacLeod
(2016))

- Chess is a game of perfect information, with a well-defined set of possible states and outcomes.
- Every board position defines the state of the game.
- Given the state, the player whose turn it is chooses a legal move to change the state of the game.
- The winner is the player who checkmates the other player's king.
- There are also rules that ensure that the game always ends, and in some cases a draw is possible.

Chess Playing is an Occupation - what do our previous models have to say about it?

- First, note that there is no uncertainty regarding the rules.
- We can study the game using the tools of information theory - specifically game theory:
 - Zermelo proved that in a game of perfect information like chess, either one player has a winning strategy or both have a drawing strategy.
 - Since the game is finite, and parties observed the state at each stage, the optimal play can be found by backwards induction from the end.
- In this case, game theory is clearly rejected by the evidence.

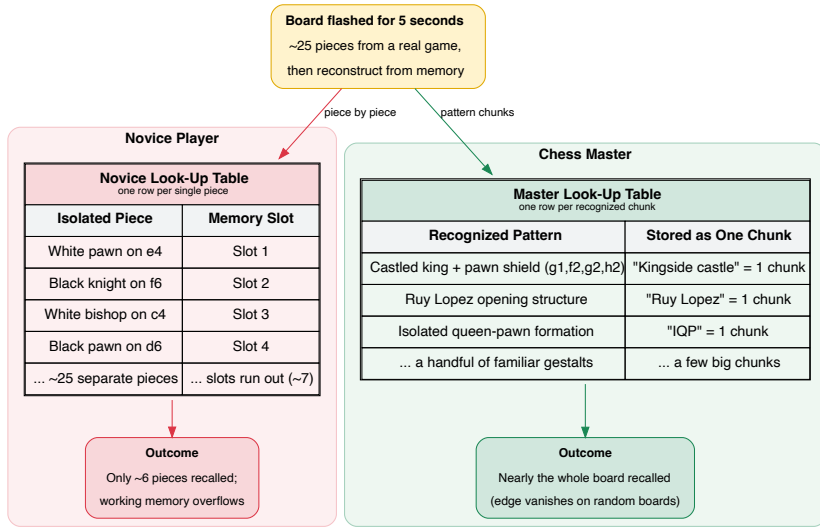
Limited time/ability to compute

So what do we know about chess skill - Chase and Simon (1973)

Same Position, Two Look-Up Tables:

Novice (square → square) vs. Master (pattern → chunk) [Chase & Simon, 1973]

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.



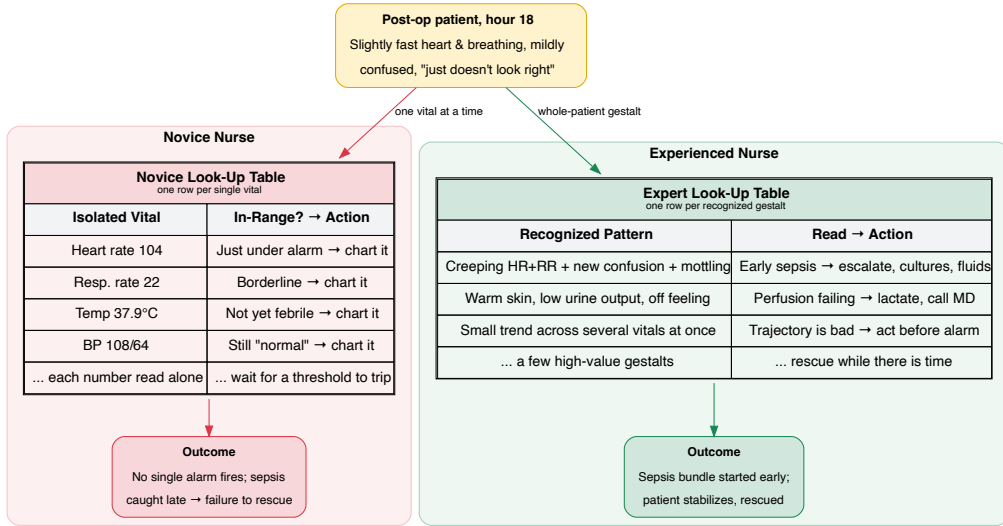
Implications

- Chess skill is a form of human capital that is **acquired** through experience and practice.
- The skill is not a function of the number of possible states, but rather the ability to recognize patterns and to recall the best moves for those patterns.
- This is a form of knowledge that is not easily codified or transmitted, but rather is acquired through learning and practice.
- This has implications for the theory of value, as it suggests that the value of a good or service is not solely determined by its physical characteristics or the preferences of consumers, but also by the knowledge and skills required to produce it.
- Churchland and Sejnowski (2016) explicitly shows that this sort of knowledge can be modelled as a lookup table.
- **MacLeod (2016)** shows that the lookup table can be viewed as human capital that increases over time.

Same Patient, Two Look-Up Tables:

Novice (vital → normal range) vs. Expert (gestalt → catch sepsis early)

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.



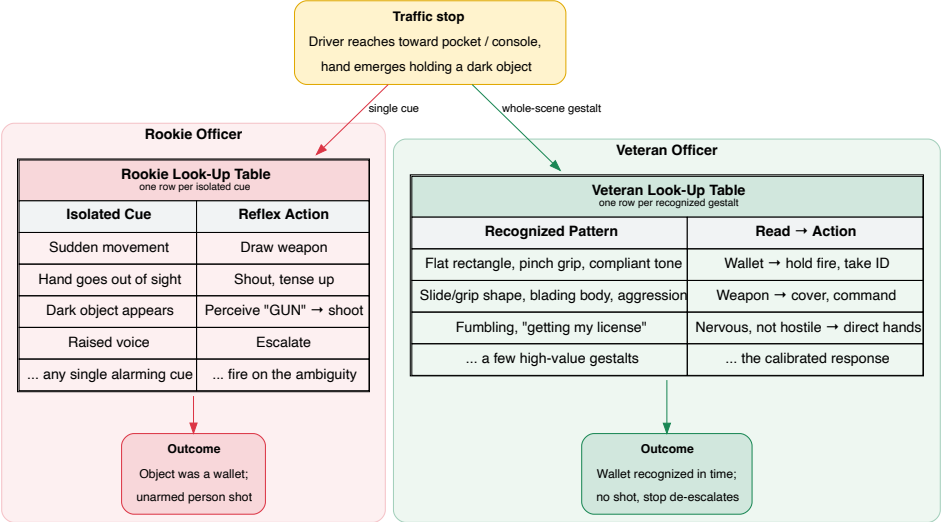
- This example illustrates that nursing skill can affect patient outcomes.
- When an adverse medical event occurs, this may lead to a lawsuit against the hospital.
- The notion of skill is relevant because the law places a higher duty of care for more skilled professionals.
Probably better to say higher 'standard' of care
- How should liability be treated in this example?

The answer of how liability is treated is provided above: a higher standard of care

Same Stop, Two Look-Up Tables:

Rookie (cue → shoot) vs. Veteran (gestalt → recognize object)

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.



- This example illustrates that police skill can affect public safety and the outcomes of law enforcement actions.
- When an incident occurs, the skill of the responding officers may determine the resolution and any legal consequences.
- The current standard for a wrongful death suit is whether or not the officer felt that their life was threatened.
- Is there a common principle at work in this and the previous case?

- One can view the current situation like our unskilled police officers - AI is the hidden gun that may be used and we are not ready to respond.
- How should we think about aligning AI with human values (“Asilomar AI Principles” (2017))?

Bounded Rationality Implies Legal Realism

The “Open Texture” of language (Hart (1958))

- Hart (1958) introduced the idea that courts are often faced with cases that fall into the penumbra of uncertainty, where the application of legal rules is not clear-cut.
- This has a number of important implications:
 - It is impossible for judges to simply “apply the law”.
 - As in the search model, but my necessary attach “human values” to their choices.
 - Let the police office, they must make a choice whether or not they are sure what is correct.

Like the police officer

- Let K represent all legal rules, a necessarily finite set that is in principle comprehensible by humans.
- Let k be a case, and $Law(K, k) \in \{\text{plaintiff, defendant}\}$ - the outcome of the case in court that must be either in favor of the plaintiff or the defendant.
- These objects can be represented with a formal language. A number of results have shown that $Law(K, k)$ will necessarily have indeterminate outcomes because of the large size of the number of possible cases (Prakken (1997) and Susskind (1987)).

These **consistent with**

- This results are consistent Hart (1958)'s view that legal realism is a necessary feature of the law.
- He illustrates this with the example of a law that states "No vehicles are allowed in the park."
- A **supervisor** who does not like children used the law to bar women with strollers. When they sue, how will the judge decide?

The Stroller Example

- An AI might correctly rule in favor on the supervisor, since strollers are vehicles. (See the movie “Colossus: The Forbin Project”)
- An human judge with experience with children and child raising would normally rule in favor of the women with strollers. **Connect experience to legal realism: Judge relies on experience because the information available and the judge’s cognitive ability is not limitless**
- As Gennaioli and Shleifer (2007) discuss, the decision by the human decision maker will become a legal precedent, whether it is efficient or not.

How are judges to resolve such cases?

- The judge must consider not only the formal rules but also the broader context, including human values and societal norms.
- The example illustrates the contrasting values of the park supervisor and the woman. In deciding the case the judge must weigh these competing set of values and interests to arrive at a fair and just decision.
- It is reasonable to suppose that strollers would be allowed.
- To arrive this we would have to rely upon the judge have “good” human values.

How are are “human values” to be incorporated in practice?

- The legal system can be viewed as an (imperfect) implementation of human values in practice.
- The law provide imperfect guidance supplemented by judicial interpretation and discretion.
- In other word in addition to the law, society has a system for selecting and appointing judges to whom we have entrusted the responsibility of interpreting and applying the law in a manner consistent with human values.

What are “good values”?

- From a law and economics perspective, we can identify “good values” as those that are implemented by **well-functioning** legal systems.
- Empirically, we observe a great deal of variation in systems and laws across the planet, and from these experiments we might identify systems that implement good **values**.

Delete “10)” and “11)”

- 10) Value Alignment: Highly autonomous AI systems should be designed so that their goals and behaviors can be assured to align with human values throughout their operation. 11) Human Values: AI systems should be designed and operated so as to be compatible with ideals of human dignity, rights, freedoms, and cultural diversity.

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Implications 2

“Asilomar AI Principles.” 2017. Future of Life Institute, August 11.

Chakravarty, Surajeet, and W. Bentley MacLeod. 2009. “Contracting in the Shadow of the Law.” *Rand Journal of Economics* 40 (3): 533–57.

Chase, William G., and Herbert A. Simon. 1973. “Perception in Chess.” *Cognitive Psychology* 4 (1): 55–81. [https://doi.org/10.1016/0010-0285\(73\)90004-2](https://doi.org/10.1016/0010-0285(73)90004-2).

Churchland, Patricia S., and Terrence J. Sejnowski. 2016. *The Computational Brain*. The MIT Press. <https://doi.org/10.7551/mitpress/11207.001.0001>.

- Cooter, Robert, and Thomas Ulen, eds. 2016. *Law and Economics, 6th Edition*. Addison-Wesley.
- Gennaioli, Nicola, and Andrei Shleifer. 2007. "The Evolution of Common Law." *Journal of Political Economy* 115 (1): 43–68. <https://doi.org/10.1086/511996>.
- Hart, H. L. A. 1958. "Positivism and the Separation of Law and Morals." *Harvard Law Review* 71 (4): 593–629. <https://doi.org/10.2307/1338225>.
- Listokin, Yair. 2010. "Bayesian Contractual Interpretation." *The Journal of Legal Studies* 39 (2): 359–74. <https://doi.org/10.1086/652459>.

- MacLeod, W Bentley. 2016. “Human Capital: Linking Behavior to Rational Choice via Dual Process Theory.” *Labour Economics* 41: 20–31. <https://doi.org/10.1016/j.labeco.2016.05.019>.
- MacLeod, W. Bentley. 2022. *Advanced Microeconomics for Contract, Institutional and Organizational Economics*. MIT Press.
- Milgrom, Paul, and John Roberts. 1992. *Economics, Organizations and Management*. Prentice Hall.
- Peierls, R. E. 1960. “Wolfgang Ernst Pauli. 1900–1958.” *Biographical Memoirs of Fellows of the Royal Society* 5: 175–92.
- Posner, Richard A. 1973. *Economic Analysis of Law*. 1st ed. Little, Brown.

Posner, Richard A. 2014. *Economic Analysis of Law*. 9th ed. Wolters Kluwer Law & Business.

Prakken, Henry. 1997. *Logical Tools for Modelling Legal Argument: A Study of Defeasible Reasoning*. Kluwer Academic Publishers. <https://doi.org/10.1007/978-94-015-8938-3>.

Savage, Leonard J. 1972 (first published 1954). *The Foundations of Statistics*. Dover Publications.

Shavell, Steven. 2004. *Foundations of Economic Analysis of Law*. Harvard University Press. <https://doi.org/10.2307/j.ctv1m0kjr4>.

Simon, Herbert A. 1955. "A Behavioral Model of Rational Choice." *The Quarterly Journal of Economics* 69 (1): 99–118. <https://doi.org/10.2307/1884852>.

Susskind, Richard. 1987. *Expert Systems in Law: A Jurisprudential Inquiry*. Clarendon Press.

Vincent, George E. 1904. "The Laws of Hammurabi." *American Journal of Sociology* 9 (6): 737–54.

Williamson, Oliver E. 1985. *The Economic Institutions of Capitalism*. The Free Press.